Unified Computational Benchmark for First-Passage Problems Luca / Generating-Function Family vs. Time-Domain Recursion Family

valley-k-small automated report

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Abstract

This document is the single active computational-method comparison report in the repository. It keeps the scientific reports separate, but collapses the comparison line into one bilingual benchmark with six workloads. The benchmark rule is *practical native-task fairness*: each report is timed on the task it actually needed to complete, rather than on an artificially synchronized full-tail horizon. Under the diagnostic protocol the median ratio $t_{\text{time}}/t_{\text{Luca}}$ is 0.0866; time recursion wins on ENC-FIXED, ENC-ANY, TT-C1, REF-S0, while Luca/GF wins on RING-1T-paper, TT-LF1. Under the curve protocol the median ratio is 0.0671; time recursion wins on ENC-FIXED, ENC-ANY, TT-C1, REF-S0, while Luca/GF wins on RING-1T-paper, TT-LF1.

1 Goal and Comparison Rule

We compare only two families.

- **Luca / generating-function family**: closed-form propagators, defect-reduced resolvent recovery, renewal in transform space, and AW/Cauchy-FFT inversion.
- **Time-domain recursion family**: exact propagation on the transient state space, including sparse absorbing recursion and absorbed pair-distribution updates.

The main benchmark is not a forced full-tail comparison. It is a report-production comparison: how much time does each family need to deliver the quantity that the scientific report actually required?

2 Unified Notation and Runtime Protocol

Let X_t be a discrete-time Markov chain, T a target set, $\tau_T = \inf\{t \geq 1 : X_t \in T\}$ the first-passage time,

$$f_T(t) = \mathbb{P}(\tau_T = t), \tag{1}$$

$$S_T(t) = \mathbb{P}(\tau_T > t), \tag{2}$$

$$\widetilde{f}_T(z) = \sum_{t\geq 1} f_T(t) z^t, \tag{3}$$

$$\widetilde{P}_{ij}(z) = \sum_{t\geq 0} \mathbb{P}_i(X_t = j) z^t = \left[(I - zP)^{-1}\right]_{ij}. \tag{4}$$

For single-target problems, renewal closes immediately:

$$\widetilde{f}_{i\to m}(z) = \frac{\widetilde{P}_{im}(z)}{\widetilde{P}_{mm}(z)}. \tag{5}$$

For multiple targets, renewal becomes a small linear system on target-to-target propagators. All timings use single-thread BLAS, one warm-up run, three measured runs, and the median wall time as the reported benchmark value.

3 Luca / Generating-Function Family: Mathematical Spine

The common transform-domain workflow is:

1. Build a defect-free propagator or a closed-form single-walker PGF.
2. Recover the defected propagator through a finite-dimensional Woodbury or determinant system.
3. Close a one-target or multi-target renewal relation to obtain the FPT generating function.
4. Invert the PGF numerically with an AW/Cauchy-FFT contour.

The inversion step evaluates

$$f(t) = [z^t]\widetilde{f}(z) = \frac{1}{2\pi i}\oint_{|z|=r} \frac{\widetilde{f}(z)}{z^{t+1}}\,dz \approx \frac{r^{-t}}{m}\sum_{k=0}^{m-1} \widetilde{f}\Big(re^{2\pi ik/m}\Big)\,e^{-2\pi ikt/m}. \tag{6}$$

This is the shared numerical shell behind the ring, encounter, two-target, and reflecting GF workloads.

3.1 Encounter-Specific GF Route

For the two-walker shortcut encounter, the defect-free pair propagator is

$$G_0\big((n_0, m_0) \to (x, y); z\big) = \frac{1}{N^2}\sum_{k_1,k_2} \frac{h_{k_1}(x, n_0)h_{k_2}(y, m_0)}{1 - z\lambda_{k_1}^{(1)}\lambda_{k_2}^{(2)}}. \tag{7}$$

The shortcut becomes a line defect on the pair torus and is restored by

$$G(z) = G_0(z) + z\,G_0(z)U\left(I - zV^\top G_0(z)U\right)^{-1} V^\top G_0(z). \tag{8}$$

A1 and A8 are therefore not the production solver for the encounter report; they are the spectral anchor from which the new timed GF route is constructed.

4 Time-Domain Recursion Family: Mathematical Spine

The time-domain family works directly with absorbing transient updates. If Q is the transient transition matrix and r the transient-to-target flux vector, then

$$u_{t+1} = u_t Q, \tag{9}$$
$$f(t+1) = u_t r, \tag{10}$$
$$S(t+1) = u_{t+1}\mathbf{1}. \tag{11}$$

For two-target problems, r becomes a two-column target flux block. For the pair encounter, the exact recursion is naturally written in matrix form:

$$J_{t+1} = P_1^\top J_t P_2, \tag{12}$$
$$f_{\text{enc}}(t+1) = \operatorname{tr}(J_{t+1}), \tag{13}$$

with the diagonal or designated encounter site zeroed after each step to enforce absorption. No asymptotic approximation is used here; the family name refers only to the domain in which the update is carried out.

5 Workload Instantiations and Configuration Figures

Each workload is now documented inside this unified report, with its own detailed configuration figure and an explicit formula-level specialization of the two method families.

workload	model	states	defects	targets
RING-1T-paper	ring_single_target	100	1	1
ENC-FIXED	ring_encounter_fixed	900	0	1
ENC-ANY	ring_encounter_any	10201	101	101
TT-C1	grid2d_two_target	1679	632	2
TT-LF1	grid2d_two_target	1679	2	2
REF-S0	grid2d_reflecting	840	9	1

Table 1: The six benchmark workloads kept in the unified comparison.

5.1 RING-1T-paper

This workload is the analytic single-target ring benchmark. The resolvent family starts from the closed-form single-walker propagator

$$\widetilde{P}_{n_0\to n}(u) = \sum_{k=0}^{N-1} \frac{h_k(n,n_0)}{1-u\alpha_k}, \tag{14}$$

$$\widetilde{F}_{n_0\to m}(u) = \frac{\widetilde{P}_{n_0\to m}(u)}{\widetilde{P}_{m\to m}(u)}, \tag{15}$$

followed by an AW/Cauchy inversion on a radius-r circle. The time solver instead iterates the absorbing chain directly. Direct speed mapping for the configuration figure: the diagnostic task is faster with Luca/GF (time=0.007163s, luca=0.0006995s, 10.2x separation). The curve task keeps the same winner (time=0.01769s, luca=0.002326s).

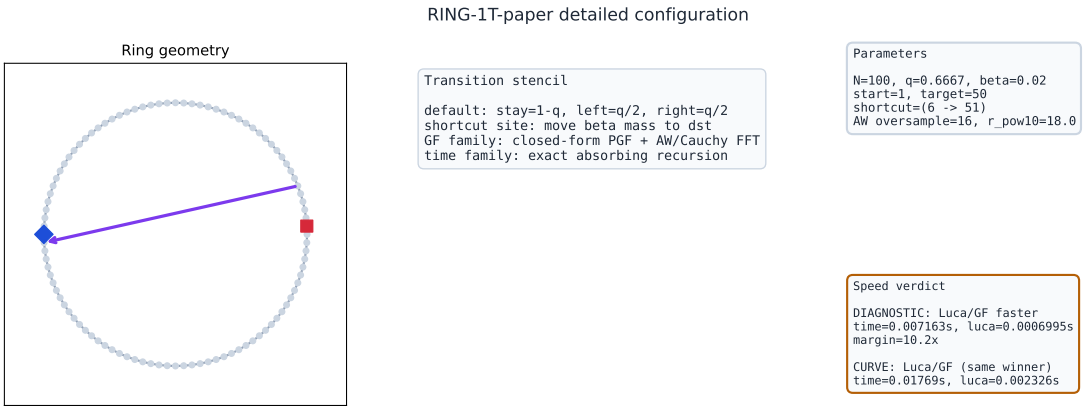

Figure 1: Detailed ring geometry for the paper-like single-target shortcut benchmark.

5.2 ENC-FIXED

The fixed-site encounter workload is a single-target first-passage problem on the pair chain. The GF family evaluates the defect-free pair propagator and closes a one-target renewal equation:

$$G_0\big((n_0, m_0) \to (x, y); z\big) = \frac{1}{N^2} \sum_{k_1,k_2} \frac{h_{k_1}(x, n_0) h_{k_2}(y, m_0)}{1 - z\lambda_{k_1}^{(1)} \lambda_{k_2}^{(2)}}, \tag{16}$$

$$\widetilde{f}_\delta(z) = \frac{G\big((n_0, m_0) \to (\delta, \delta); z\big)}{G\big((\delta, \delta) \to (\delta, \delta); z\big)}. \tag{17}$$

The time family evolves the absorbed pair density exactly on the N^2 state space. Direct speed mapping for the configuration figure: the diagnostic task is faster with Time recursion (time=0.006232s, luca=0.3488s, 56.0x separation). The curve task keeps the same winner (time=0.005959s, luca=0.3508s).

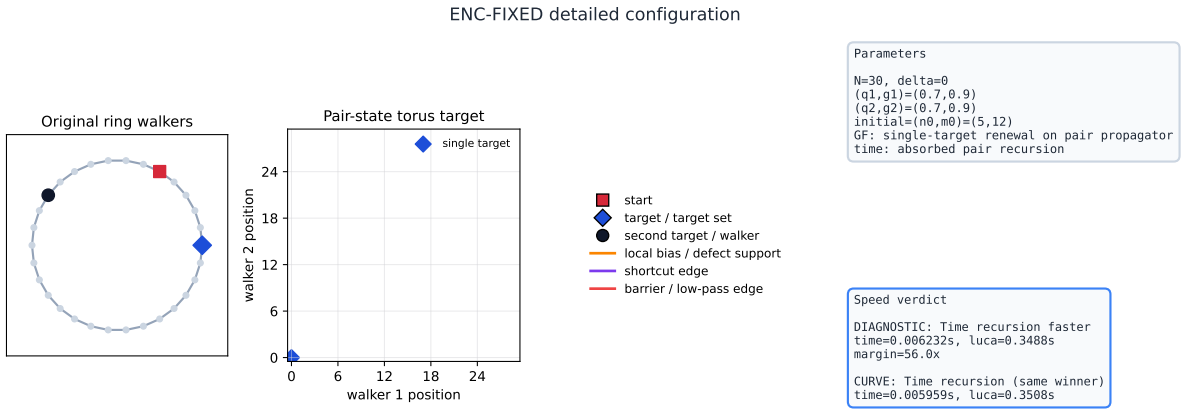

Figure 2: Detailed configuration for the fixed-site encounter benchmark, including the single pair-state target.

5.3 ENC-ANY

The anywhere-encounter workload promotes the target from one pair-state to the full diagonal set $D = \{(x, x) : x \in \mathbb{Z}_N\}$. The GF family uses the same defect-free pair propagator but now closes a multi-target renewal system:

$$G(z) = G_0(z) + z\, G_0(z) U \left(I - zV^\top G_0(z) U\right)^{-1} V^\top G_0(z), \tag{18}$$

$$\widetilde{\mathbf{f}}_D(z) = G_{DD}(z)^{-1} G_{Ds}(z), \tag{19}$$

$$\widetilde{f}_{\mathrm{enc}}(z) = \mathbf{1}^\top \widetilde{\mathbf{f}}_D(z). \tag{20}$$

Here the shortcut acts as a line defect on the pair torus, so the Woodbury correction is the bridge between the defect-free pair spectrum and the actual encounter PGF. Direct speed mapping for the configuration figure: the diagnostic task is faster with Time recursion (time=0.0123s, luca=2.317s, 188.3x separation). The curve task keeps the same winner (time=0.02134s, luca=4.664s).

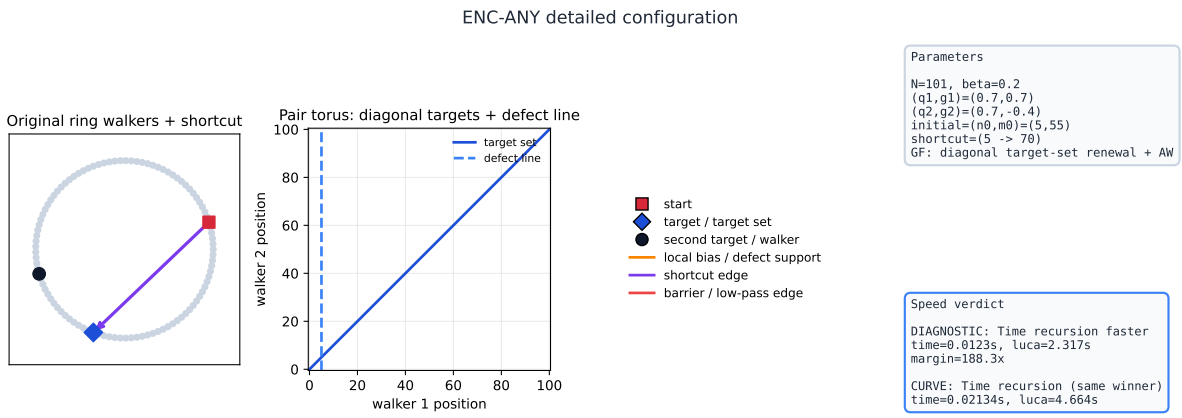

Figure 3: Detailed configuration for the anywhere-encounter benchmark, including the diagonal target set and the shortcut-induced defect line.

5.4 TT-C1

C1 is the practical two-target corridor benchmark. The GF family uses a defect-reduced resolvent instead of a full dense transient solve:

$$Q = Q_0 + U\Delta V^\top, \tag{21}$$

$$(I - zQ)^{-1} = (I - zQ_0)^{-1} + z(I - zQ_0)^{-1}U\left(I - zV^\top(I - zQ_0)^{-1}U\Delta\right)^{-1}V^\top(I - zQ_0)^{-1}, \tag{22}$$

$$\widetilde{\mathbf{f}}(z) = G_{TT}(z)^{-1}G_{Ts}(z), \tag{23}$$

where $T = \{m_1, m_2\}$. The time solver is the sparse exact absorbing recursion on the transient graph, continued to the long horizon actually needed by the scientific claim. Direct speed mapping for the configuration figure: the diagnostic task is faster with Time recursion (time=0.01313s, luca=41.19s, 3137.2x separation). The curve task keeps the same winner (time=0.06642s, luca=41.03s).

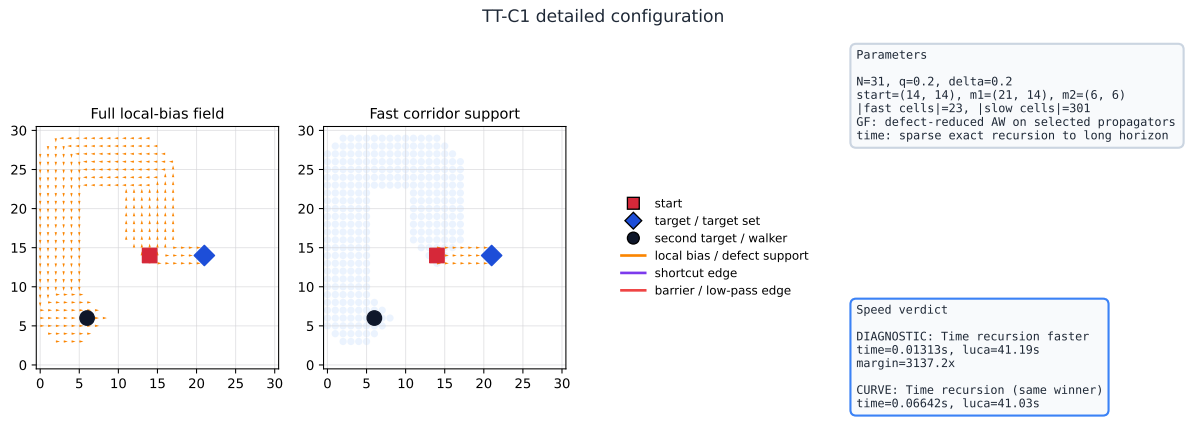

Figure 4: Detailed C1 geometry, showing both the full local-bias field and the corridor support that determines the defect-reduced dimension.

5.5 TT-LF1

TT-LF1 keeps the same two-target mathematics but collapses the defect support to an ultra-sparse local perturbation. This is the in-repo positive anchor where the defect-reduced GF route can beat long-horizon sparse recursion. Direct speed mapping for the configuration figure: the diagnostic task is faster with Luca/GF (time=0.1901s, luca=0.01725s, 11.0x separation). The curve task keeps the same winner (time=0.1904s, luca=0.01719s).

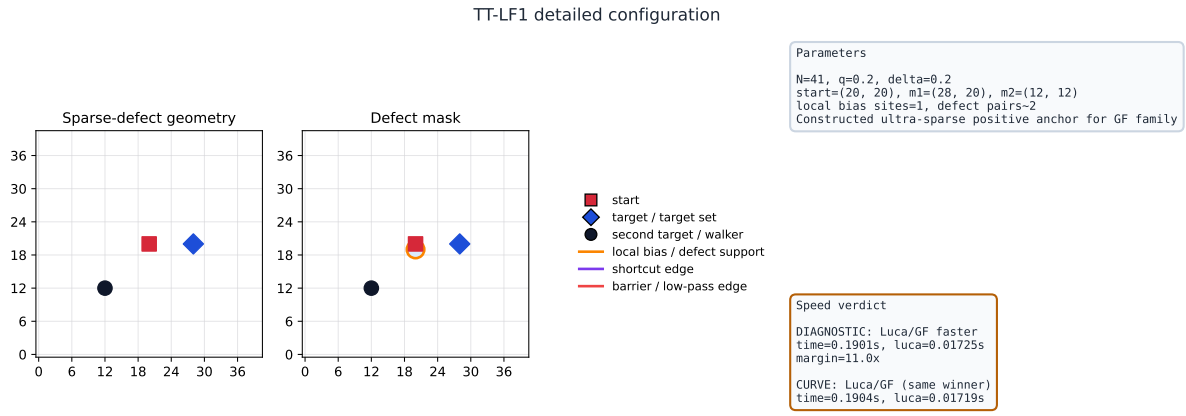

Figure 5: Detailed LF1 sparse-defect geometry used as the positive Luca/GF anchor.

5.6 REF-S0

REF-S0 is the reflecting low-defect control. The GF family remains a single-target resolvent solve, but here the full AW route is still numerically feasible because the defect count is deliberately kept small:

$$\widetilde{F}_{s\to m}(z) = \frac{G_{sm}(z)}{G_{mm}(z)}, \tag{24}$$

$$f(t) = [z^t]\widetilde{F}_{s\to m}(z). \tag{25}$$

The time family is the standard exact absorbing recursion on the reflecting lattice. Direct speed mapping for the configuration figure: the diagnostic task is faster with Time recursion (time=0.006363s, luca=0.04098s, 6.4x separation). The curve task keeps the same winner (time=0.01758s, luca=0.15s).

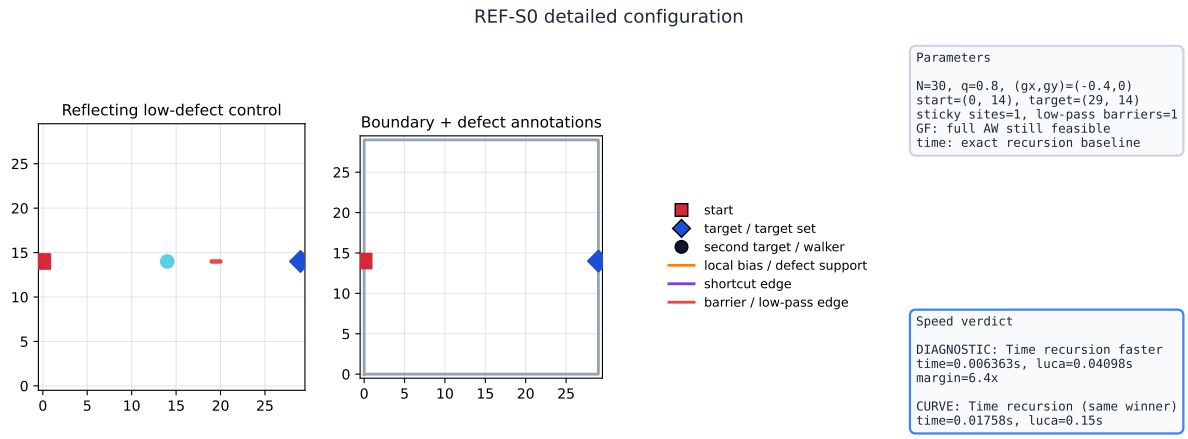

Figure 6: Detailed reflecting low-defect S0 control geometry, highlighting the sticky site and the partially permeable barrier.

6 Main Benchmark Results

workload family	Luca/GF family	time recursion family	comment
RING-1T-paper	$O(mN) + O(m \log m)$	$O(TN)$	closed-form single-target PGF plus AW
ENC-FIXED	$O(mN^3)$	$O(TN^3)$	single-target renewal on the pair chain
ENC-ANY	$O(mN^3)$	$O(TN^3)$	diagonal target set with $\lvert D\rvert = N$
TT-C1 / TT-LF1	$O(mM^3) + O(m \log m)$	$O(T\lvert E\rvert)$	defect-reduced AW vs sparse exact recursion
REF-S0	$O(mn_T^3) + O(m \log m)$	$O(T\lvert E\rvert)$	low-defect full AW vs exact recursion

Table 2: Family-level complexity summary.

workload	source report	Luca/GF solver	time solver
RING-1T-paper	ring_lazy_flux	ring_analytic_aw	ring_time_absorption
ENC-FIXED	ring_two_walker_encounter_shortcut	encounter_fixedsite_gf_aw	pair_fixedsite_time_recursion
ENC-ANY	ring_two_walker_encounter_shortcut	encounter_anywhere_gf_aw	pair_time_recursion
TT-C1	grid2d_two_target_double_peak	two_target_defect_reduced_aw	two_target_sparse_exact
TT-LF1	grid2d_two_target_double_peak	two_target_defect_reduced_aw	two_target_sparse_exact
REF-S0	grid2d_reflecting_bimodality	reflecting_full_aw	reflecting_exact_recursion

Table 3: Family-internal solver mapping for the six workloads.

One-glance overview: the numbered tiles 1-6 match the numbered workload groups on the runtime chart

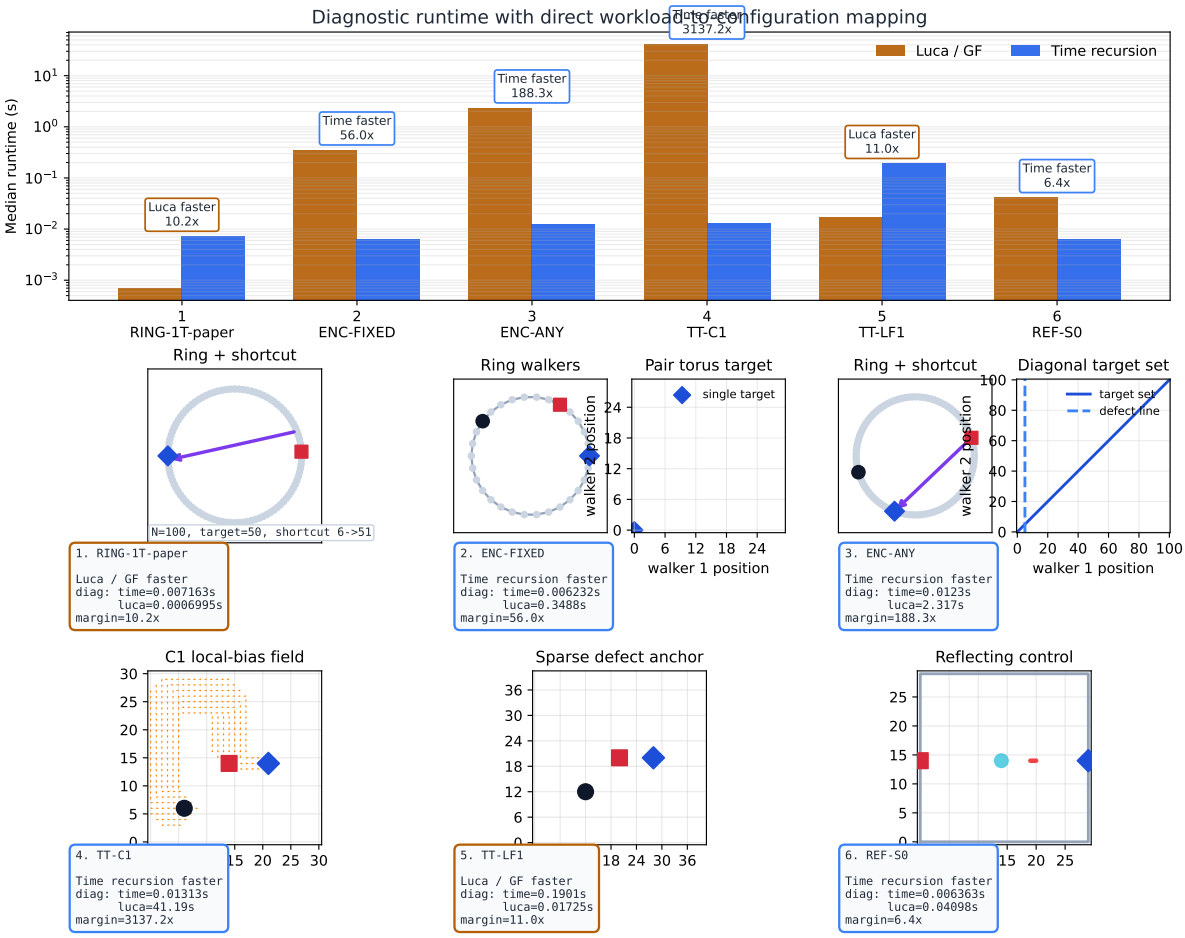

Figure 7: One-glance benchmark overview. The numbered mini-panels 1–6 are the six workload configurations, and the numbered groups on the top runtime chart follow the same order. This is the main practical benchmark figure used in the report.

workload	Luca/GF (s)	time recursion (s)	effective T (L/T)	recommended	L^1
RING-1T-paper	0.0006995	0.007163	800/800	Luca / generating-function family	1.643e-06
ENC-FIXED	0.3488	0.006232	900/900	Time-domain recursion family	1.382e-10
ENC-ANY	2.317	0.0123	420/420	Time-domain recursion family	8.994e-14
TT-C1	41.19	0.01313	800/1200	Time-domain recursion family	9.912e-10
TT-LF1	0.01725	0.1901	80/12000	Luca / generating-function family	2.135e-10
REF-S0	0.04098	0.006363	300/300	Time-domain recursion family	1.733e-09

Table 4: Diagnostic-task runtime table.

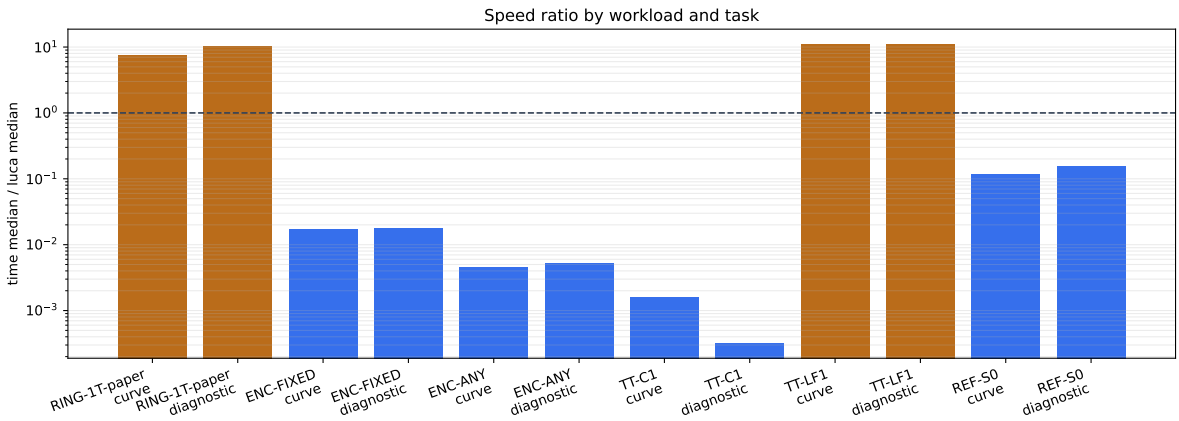

Figure 8: Per-workload ratio $t_{\text{time}}/t_{\text{Luca}}$. Ratios above one favor the Luca/GF family.

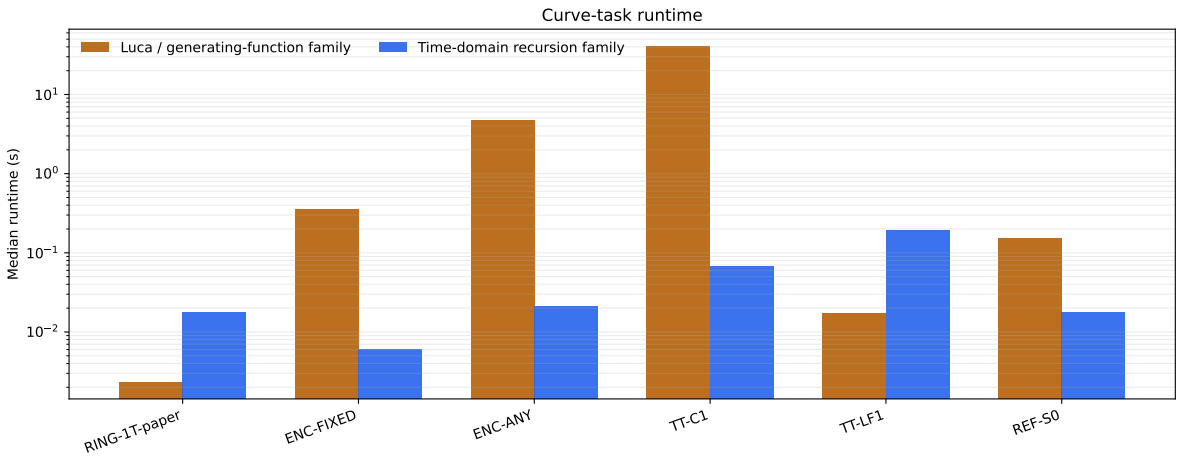

Figure 9: Curve-task runtime, kept secondary so that long-tail costs do not overwrite the practical report-production benchmark.

workload	Luca/GF (s)	time recursion (s)	effective T (L/T)	recommended	L^1
RING-1T-paper	0.002326	0.01769	2000/2000	Luca / generating-function family	7.588e-10
ENC-FIXED	0.3508	0.005959	900/900	Time-domain recursion family	1.382e-10
ENC-ANY	4.664	0.02134	800/800	Time-domain recursion family	1.046e-13
TT-C1	41.03	0.06642	800/6000	Time-domain recursion family	9.912e-10
TT-LF1	0.01719	0.1904	80/12000	Luca / generating-function family	2.135e-10
REF-S0	0.15	0.01758	1200/1200	Time-domain recursion family	3.051e-09

Table 5: Curve-task runtime table.

7 Recommendation Matrix

Scenario	Recommendation
Single-target ring with analytic PGF	Prefer Luca/GF when the closed form already exists and longer windows are needed.
Shortcut encounter (anywhere/fixed-site)	Prefer time recursion for production figures; use GF when transform-domain renewal validation is itself the task.
2D two-target, medium/high defect	Prefer time recursion; C1-like cases still punish the defect-reduced AW route.
2D two-target, ultra-sparse defect	Luca/GF can win; TT-LF1 is the in-repo positive anchor.
2D reflecting low-defect control	Time recursion still wins on wall time in this repo.

Table 6: Decision matrix derived from the six-workload benchmark.

8 Conclusion

The comparison line is now fully centralized in one report. The scientific reports remain focused on science, while this document owns the algorithmic comparison, the fairness rule, the workload inventory, and the implementation-level guidance for choosing between transform-domain and time-domain solvers.

A Appendix A: Symbol Table and Variable Mapping

$$P : \text{full transition operator}, \qquad Q : \text{transient transition operator}, \tag{26}$$

$$r : \text{transient-to-target flux}, \qquad U\Delta V^\top : \text{defect update}, \tag{27}$$

$$G_0(z) : (I - zQ_0)^{-1}, \qquad G(z) : (I - zQ)^{-1}, \tag{28}$$

$$m : \text{AW FFT length}, \qquad r : \text{AW contour radius}. \tag{29}$$

B Appendix B: Detailed GF Derivations

The GF family reduces every workload to the same algebraic template. Starting from a defect-free Q_0, write

$$Q = Q_0 + U\Delta V^\top, \tag{30}$$

$$I - zQ = (I - zQ_0)\Big[I - z(I - zQ_0)^{-1}U\Delta V^\top\Big], \tag{31}$$

$$(I - zQ)^{-1} = (I - zQ_0)^{-1} + z(I - zQ_0)^{-1}U\Big[I - zV^\top(I - zQ_0)^{-1}U\Delta\Big]^{-1}V^\top(I - zQ_0)^{-1}. \tag{32}$$

For a target set $T = \{m_1, \ldots, m_K\}$, renewal is written as

$$G_{sT}(z) = \widetilde{\mathbf{f}}_T(z)^\top G_{TT}(z), \tag{33}$$

$$\widetilde{\mathbf{f}}_T(z) = G_{TT}(z)^{-1}G_{Ts}(z). \tag{34}$$

The one-target formula is the $K = 1$ specialization of this matrix equation.

C Appendix C: Detailed Time-Recursion Derivations

Given the transient state vector u_t, exact recursion uses

$$u_0 = e_s, \tag{35}$$

$$u_{t+1} = u_t Q, \tag{36}$$

$$f(t+1) = u_t r, \tag{37}$$

$$S(t+1) = u_{t+1}\mathbf{1}. \tag{38}$$

For two-target problems, let $R = [r^{(1)}\ r^{(2)}]$ and then $f_1(t+1) = u_t r^{(1)}$, $f_2(t+1) = u_t r^{(2)}$. For pair encounters, the matrix recursion form is numerically convenient because the encounter condition is a structured absorbing set on the pair torus.

D Appendix D: Per-Workload Formula Specialization

RING-1T-paper uses a scalar renewal. ENC-FIXED uses a single-target renewal on the pair chain. ENC-ANY uses a diagonal target-set renewal on the pair torus. TT-C1 and TT-LF1 use a two-target renewal after defect-reduced selected-propagator recovery. REF-S0 uses the single-target reflecting resolvent. These are exactly the formulas instantiated in Section 5 alongside the detailed configuration figures.

E Appendix E: Complexity, Memory, Error Metrics, and Benchmark Protocol

The runtime comparison is accepted only after a same-window numerical agreement check between the two families. For each workload, the overlap window is reported through L^1, L^∞,

and a peak-location consistency flag. This means wall time is never reported without a paired numerical sanity check.

F Appendix F: Embedded Historical Full-FPT Note

The repository no longer keeps an external active compare report for the old fixed full-tail fairness question. Instead, the historical note is embedded here so that the active comparison line remains single-source.

quantity	value
diagnostic median t_{time}/t_{Luca}	0.08657
curve median t_{time}/t_{Luca}	0.06708
historical policy retained here	embedded appendix only
external compare report	retired

Table 7: Embedded historical fairness note retained inside the unified report.